

Psychology and Neuroscience 41st Annual Graham Goddard In-House Conference



April 30-May 1, 2015
Life Sciences Centre
Room 4258/4263

2015 Psychology and Neuroscience In-House Conference

The 2015 Psychology and Neuroscience Graham Goddard In-House Conference

Welcome! We would like to thank you for attending the 41st Annual Graham Goddard In-House Conference. This is an excellent opportunity to hear about the ongoing research of students, faculty and colleagues. It is your participation that makes this event a success.

Thanks should also be extended to the In-House Conference Committee. The time that the following individuals put into the planning and organization of the In-House is greatly appreciated: Sophie Jacques, Ivy-Lee Kehayes, Jill King, Suzanne King, Lynn MacKenzie, R.J. Redden, Michelle Tougas and Francesco Usai

Dr. Graham V. Goddard

Dr. Graham V. Goddard (1938-1987) was a leading researcher at Dalhousie University. Goddard invented the kindling model of epilepsy that has, since its introduction in the 1960s, become widely used research tool in studies of long-term potentiation (LTP) and epilepsy. His work was published in Nature, Science, along with other high profile journals. During his time at Dalhousie, Goddard helped establish Dalhousie as a leading research institution. Goddard also organized the annual Psychology Department In-House Conference, to help investigators in the department update others about their work and encourage academic dialogue. It is in his honour that we continue this tradition today.

Friendly Reminders

Talks will be held in LSC 4253/4268.

Coffee and tea will be available in the 4th Floor Study Centre (4316).

Lunches will take place in the 2nd Floor Lounge.

There will be a dinner and reception to follow at the end of Friday May 1st, in the 2nd Floor Lounge.

Thursday, April 30, 2015 (LSC 4263)

9:00–10:30am: Session 1. Chair: Ray Klein

9:00 T1) Introduction and History of InHouse

Raymond M. Klein

9:15 T2) Understanding second language in the brain with a mini-language model

Kaitlyn M. Tagarelli, Aaron J. Newman, Michael T. Ullman

9:30 T3) Brain Activation Associated with American Sign Language Learning

Emily Patrick, Emily McGuire, Aaron Newman

9:45 T4) Visualize your imagination

Sreejith P G, Graham Flick, Antoine Tremblay, Aaron J Newman

10:00 T5) The Effect of Cueing on the Directed Forgetting Effect

Laura Cutmore & Tracy Taylor-Helmick

10:15 T6) The evolution of persons

John Barresi

10:30–11:00am: COFFEE BREAK (LSC 4316)

11:00–12:15pm: Session 2. Chair: Tracy Taylor-Helmick

11:00 T7) Individual differences in attentional control: The case of action video game players

Joseph Chisholm

11:15 T8) Effects of Sleep Restriction on Components of Attention: Interim Results

Jasmyn E. A. Cunningham, Stephanie A. H. Jones, Joshua P. Salmon, Franziska Kintzel, Gail A. Eskes, Benjamin Rusak

11:30 T9) Finding physiological indices of alertness in healthy adults using electroencephalography (EEG)

Yoko Ishigami, Jason Satel, Shaun Boe, Raymond M. Klein, & Gail Eskes

11:45 T10) Directed Forgetting: Consequences of a Forget Instruction

Andrew Beattie, Kate Thompson & Tracy Taylor-Helmick

12:00 T11) From gestures to gaming: Visible embodiment of one's cognitive state

Joseph Chisholm

12:15–1:30pm: LUNCH AND POSTER SESSION (2nd Floor Lounge)

P1) Looking at endogenous prior entry effects in baseball judgements at first base

Ghislain d'Entremont, RJ Redden, Mike Lawrence & Raymond Klein

Poster 2 has been moved to Friday

P2) Exploring sleep and circadian rhythm disturbance in an animal model of early stage Alzheimer's disease

Roxanne Sterniczuk, Setareh Lahasee & Ben Rusak

P3) Investigation of glial fibrillary acidic protein (GFAP) in the primary visual cortex during normal development and in the absence of visually-driven activity.

NESC3370: R. Kindellan, J. Morrow, A. Shahisavandi, H. Thorpe, K. Trotman, A. Hidayat, M. Barbaro, J. Blacklaws, A. Brown, J. Greenan, S. Kwiatkowski, L. Meisner, A. Pieterse, S. Shehzaad, M. Taylor, JA Horne, Kevin Duffy

P4) Age related changes in anxiety, motor performance and spatial learning in the 5xFAD mouse model of Alzheimer's Disease.

Emre Fertan and Dr. Richard E. Brown

1:30–2:45pm: Session 3. Chair: Stephanie Jones

1:30 T12) The Mediating Role of Gambling Motives in Explaining the Prospective Relations of Personality to Gambling Outcomes in Emerging Adults.

Stewart, S. H., Mackinnon, S. P., Lambe, L., & Pearson, R.

1:45 T13) Don't Stumble Tumble: A Videogame to Reduce the Use of General Anesthesia in Pediatric Magnetic Resonance Imaging

Melissa Howlett & Jill Chorney

2:00 T14) Real Time Neurofeedback During Imagined Imitation of Arm Movements: A Pilot Study

Christopher Friesen & Shaun Boe

2:15 T15) Exposure to tobacco-related cues does not enhance cravings in patients with psychotic disorders relative to control smokers

Marie-Eve Couture, Sean P. Barrett, Hera Schlagintweit, Phil Tibbo, Sherry H. Stewart, Heather Milliken, & Kimberley P. Good

2:30 T16) A pilot study to examine the effects of computerized working memory training on working memory and attention in people with Parkinson's disease

Stephanie A. H. Jones, Elaine Skene, Janet Green, Gail A. Eskes

2:45—3:15pm: COFFEE BREAK (LSC 4316)

3:15–4:30pm: Session 4. Chair: Aaron Newman

3:15 T17) Building a cortical network of semantic concepts in a second language: ERP evidence of fast learning via computerized games

Kiera O'Neil, Aurélie Lagarrigue, Aaron J. Newman, Cheryl Frenck-Mestre

3:30 T18) Virtual reality environments for aphasia rehabilitation

Francesco Usai & Aaron J. Newman

3:45 T19) Diffusion Tensor Magnetic Resonance Imaging and Assessment of Olfactory Function as Early Indicators of Parkinson's Disease and Alzheimer's Disease

Rebecca Hastings & Dr. Kim Good

4:00 T20) Inducing Spike Timing Dependent Plasticity in the Human Brain Through Paired Associative Stimulation.

Tony Ingram & Shaun Boe

4:15 T21) Sequential actions: why do upcoming perceptual - but not motor - tasks affect grasping?

Kevin LeBlanc & David Westwood

Friday, May 1, 2015 (LSC 4263)

9:00–10:30am: Session 5. Chair: Leslie Phillmore

- 9:00 T22) Is there a pharmacological intervention to replace darkness? Preliminary results from a collaborative study (M.I.T.) of the use of intraocular injection of tetrodotoxin (TTX) to recover vision in animal models of amblyopia**
Donald Mitchell, Kevin Duffy, Paige Northrup, Ming-fai Fong & Mark Bear
- 9:15 T23) Feedback content can alter ERPs and perceptual realignment during a prism adaptation task**
Stephane MacLean, Yoko Ishigami, Cameron Hassall, Olav Krigolson, Gail Eskes
- 9:30 T24) Optogenetic silencing of mouse primary visual cortex effects contrast adaptation in V1 but not the LGN.**
Jillian L. King, Kurt R. Stover, Matthew P. Lowe, Nathan A. Crowder
- 9:45 T25) Horizontal cell inhibition of cones via ATP hydrolysis**
Sunghan Jung & William H. Baldrige
- 10:00 T26) Do zebra finches speak Russian? An operant task to test discriminability of mothers' song and speech to their infants.**
Jordan Fisk, Christine Tsang, Simone Falk, Leslie Phillmore
- 10:15 T27) The dynamics of inhibition in the aftermath of a cue is contingent on reflexive oculomotor activation.**
Ralph S. Redden, Matthew D. Hilchey & Raymond M. Klein

10:30–11:00am: COFFEE BREAK (LSC 4316)

11:00–12:00pm: Session 6. Chair: Tara Perrot

- 11:00 T28) Three optical mechanisms in wing spots of a large dolichopodid fly, *Liancalus genualis*.**
Steve Shaw
- 11:15 T29) A Comparative Approach to Understanding Sickness Mediated Behavioural Changes**
Ken Sullivan
- 11:30 T30) Coping with constraints: nutritional stress elicits changes in immune function in the caterpillar, *Manduca sexta***
Kurtis Turnbull, Ilya F. Kovalko, Gillian Davies, Russell H. Easy & Shelley A. Adamo

11:45 T31) Coping with stress early in life to build resilience for later: Further evidence for the negative effects of helicopter parenting.

Tara Perrot

12:00–1:15pm: LUNCH AND POSTER SESSION (2nd floor lounge)

P5) Using the SpyTrain videogame as a means to improve visual search in the older adults

Michael Gardner & Ella Daly

P6) Examining mood, drinking motives, and alcohol use over the menstrual cycle

Megan Hodgins, Amanda Hudson, Roisin M. O'Connor, Sherry H. Stewart

P7) Anatomy and Pathology of the Human Olfactory Bulb in Alzheimer's Disease

H. Hamodat, M.K. Cash, S. Darvesh

P8) Amyloid-beta plaque load, cell density, and soma size in the 3xTg-AD mouse model of Alzheimer's disease

Kylie Duncan, Miranda Frison, Michelle Lehman, Hannah Wood

1:15–2:45pm: Session 7. Chair: Sherry Stewart

1:15 T32) A scoping review of information about child health shared over social media

Michelle E Tougas, Christine T Chambers, Penny Corkum

1:30 T33) Gender Differences in the Mutual Influences of Drinking Motives in Couples

Ivy-Lee Kehayes, Sean P. Mackinnon, Ken Leonard, Ron Fraser & Sherry H. Stewart

1:45 T34) Relationship conflict and drinking to cope: An actor-partner interdependence model

Mackinnon, S. P., Pieterse, A., Keyhayes, I., Leonard, K., Fraser, R., & Stewart, S. H.

2:00 T35) Periodic limb movements during sleep in children with Attention-Deficit/Hyperactivity Disorder: Baseline frequency and impact of psycho-stimulant medication.

Esmot A Begum, Amy Goodday, Kathy F. Spurr, Malgorzata Rajda, Penny Corkum

2:15 T36) Attentional biases in problem and non-problem gamblers

Amanda Hudson & Sherry H. Stewart

2:30 T37) Emotional Decision-Making and Psychotic-Like Experiences

Lynn MacKenzie, Rudolf Uher & Shannon Johnson

2:45-3:15pm: COFFEE BREAK (LSC 4316)

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3:15-5:00pm: Session 8. Chair: Richard Brown

3:15 T38) Why study the history of neuroscience?

Richard Brown

3:30 T39) Representing the Brain: Modeling neuroanatomy from the Early Modern to the Contemporary period

Brianna Bird

3:45 T40) Golgi versus Cajal and the Neuron Doctrine

Emre Fertan

4:00 T41) Wilder Penfield's lasting contributions to neuroscience

Ryan Wheeler

4:15 T42) I'm significant and you're not

John Christie

4:30 T43) Do children with dyslexia have deficits in spoken sentence comprehension?

Erin Robertson

5:00 Closing Comments

5:15 Dinner /Reception in the 2nd Floor Lounge

T1) History of the InHouse Conference

T2) Understanding second language in the brain with a mini-language model

Kaitlyn M. Tagarelli, Aaron J. Newman, Michael T. Ullman

How does an adult brain learn a second language (L2)? The complex nature of L2 learning – such as the amount of time it takes to learn a language, the variety of conditions in which people learn languages, and the vast individual differences between learners – has made it difficult to completely answer this question. In order to address some of these issues, research in psychology and applied linguistics has turned toward artificial linguistic systems, like word lists, artificial grammars, and artificial languages, which can be learned quickly in highly controlled settings. However, the extent to which results from this research can be generalizable to natural languages is often called into question. This presentation evaluates the utility of a more natural language model, a mini-language, and presents a study in which learners were exposed to a mini-language while continuous behavioral and fMRI measures were acquired. The results from this study support dual-mechanism models of language learning and processing, in which the neural bases of language change as a function of exposure and proficiency, with different patterns for lexical/semantics and grammar.

T3) Brain Activation Associated with American Sign Language Learning

Emily Patrick, Emily McGuire, Aaron Newman

Studying sign languages gives the opportunity to explore a language that is unique in terms of the form of communication. Neuroimaging studies have shown that signed and spoken languages engage similar neural networks in native learners of each. Experience in visual-manual language affects how people process not only linguistic signs, but non-linguistic gesture as well. It is not known, however, how much sign language knowledge is necessary for these changes in brain activation patterns to occur. In the proposed research, we will use functional MRI to investigate the changes in brain activation patterns in response to signs and gestural stimuli, as hearing English speakers learn American Sign Language (ASL). We hypothesize that when responding to ASL stimuli (and potentially gestures) patterns will shift from bilateral activation of areas responsible for biological motion and human action perception, towards left lateralized activation of networks associated with linguistic processing. This research will add to the current understanding of how ASL knowledge affects processing of not only signs but also non-linguistic gestures. In addition, by examining novice ASL learners, we will gain a better understanding of how people learn a second language in a new modality, something that has yet to be determined.

T4) Visualize your imagination

Sreejith P G, Graham Flick, Antoine Tremblay, Aaron J Newman

Communication is one the most important skills that is required to be successful in life. There are so many creative people around the world who do not have the ability to speak and are unable to showcase their talents to the full extent. Through my research I am developing a technique to project what we imagine onto a computer screen. Using this technique, people who do not have the ability to speak may eventually be able to convert what they want to convey into visuals, making communication easier for them.

My research focuses on harnessing the power of machine learning to build a system that can decode visual imagery from neural activation, with the goal of predicting what people are imagining based on their patterns of brain activity. To do this, Naïve Bayes and C-forest classification models were trained on magnetoencephalography (MEG) recordings acquired while subjects viewed a series of images, thus establishing profiles of neural activation associated with visual components of the imagination. These models were then used to predict what images the subjects saw based on their observed neural activation. Ensemble techniques in machine learning were used to make the model more general and less prone to over-fitting. Initial results indicated good success in predicting stimulus features including shape, orientation, and colour based on brain activation alone. Different brain regions contributed most to correct classification of different features, and classification accuracy was improved by using data exclusively from the temporal and occipital lobes, compared to using data from the whole brain. If these models are successful, they will constitute an initial step towards reconstructing authentic human visual experiences.

T5) The Effect of Cueing on the Directed Forgetting Effect

Laura Cutmore and Tracy Taylor-Helmick

This study explored the possible effects of cueing on the Directed Forgetting Effect (DFE). Thirty-six participants were shown words followed by a 'remember' or 'forget' instruction and then given a yes/no recognition memory test. In order to test whether cueing would decrease the DFE, the memory test was blocked into Remember Cued, Forget Cued, and Uncued sections. The proportion of corrected yes responses were calculated to find the results. The results showed a DFE as there was a main effect for memory instruction [$F(1,35)=89.780$, $MSe=.030$, $p<.001$] with Remember greater than Forget ($Mr=.5293$, $Mf=.2566$, $SEr=.04006$, $SEf=.02209$), but there was no effect of cueing [$F(1,35)=.102$, $MSe=.014$, $p>.001$] and no interaction ($F_s<1$) between memory instruction and cueing. From this study, it was found that cueing has no effect on the DFE.

T6) The evolution of persons

John Barresi

In contrast to other animals, human beings are aware of themselves as selves and persons. How and why did this level of self-consciousness come about? I argue that it came about to meet the challenge for hominins to work together cooperatively in the complex physical and social environment of the Pleistocene. Through cooperation that involved both foraging and breeding, our ancestors became cognitively and emotionally bound to each other in order to engage in temporally extended joint activities that provided a fitness advantage for all members of the collective. Individuals became aware of themselves and others equally as persons involved in a variety of roles that were potentially reversible. The agent-neutral concept of person made possible uniform calculations of costs and benefits to self and other that could be used to maintain equity across cooperative activities and provide roughly equal fitness gains for all members in the collective.

T7) Individual differences in attentional control: The case of action video game players

Joseph Chisholm

The past couple of decades have seen a marked increase in the popularity of video game playing, especially games within the *action* genre. Given how cognitively complex action video games are, coupled with the time people spend playing these games, interest has grown in how extensive experience with action video games may affect those who play them. To date, studies comparing action video game player (AVGPs) and non-player performance on a variety of cognitive tasks, many which engage aspects of selective attention, have revealed that AVGPs commonly outperform non-players. To date, the collective evidence has suggested that action video game playing provides players with greater or more flexible control over the spatial allocation of attentional resources. I will present recent work that compared AVGPs and non-player performance in attention and oculomotor capture tasks to assess whether AVGPs are able to engage such superior attentional control to better resist visual distraction.

T8) Effects of Sleep Restriction on Components of Attention: Interim Results

Jasmyn E. A. Cunningham, Stephanie A. H. Jones, Joshua P. Salmon, Franziska Kintzel, Gail A. Eskes, Benjamin Rusak

Many patients with psychiatric illnesses have both cognitive problems and disturbed sleep. These may be causally related since sleep loss is known to disrupt many aspects of cognition. Attention is a complex function that includes several separate, but interacting neural systems guiding vigilance (readiness to respond), orienting (searching for and the selection of stimuli for further processing) and executive control (attention resource allocation) abilities. However, it is not clear which of these underlying component processes of attention are most affected by sleep loss. The Dalhousie Computerized Attention Battery (DalCAB, 8 computerized reaction time tasks purported to measure aspects of vigilance, orienting and executive control) was administered on two occasions (within subjects design) to healthy participants (women aged 19-25 years), once after a nine hour overnight sleep opportunity and a second time after a three hour overnight sleep opportunity (sleep deprivation condition). Self-ratings of sleepiness and mood states were also obtained following each sleep condition. Interim results evaluating which attentional systems are affected by acute sleep loss will be reported.

T9) Finding physiological indices of alertness in healthy adults using electroencephalography (EEG)

Yoko Ishigami, Jason Satel, Shaun Boe, Raymond M. Klein, and Gail Eskes

The alerting network plays a special role in the systems of attention. In our previous work, we determined that the alerting network interacts both with the orienting and executive networks of attention. However, these interactions depends on the modality of alerting (visual and auditory) and age. The goal of the current study is to examine the physiological indices of the alerting effects previously identified in an identification task as a function of alerting cue modality. The task for the participants (n=15/ modality condition, mean age = 20.8 yrs) is to indicate the direction of an arrow in the middle of the screen following a visual or auditory alerting cue in half of the trials while electroencephalography was recorded. Cue-elicited event-related potentials (ERPs), including the P1 and P300 ERP components, will be examined for each modality. Details of the relationship between alerting effects in ERPs and behaviour will be discussed.

T10) Directed Forgetting: Consequences of a Forget Instruction

Andrew Beattie, Kate Thompson and Tracy Taylor-Helmick

This study aimed at replicating the Directed Forgetting Effect through the use of videos of people saying a word. Typically, research has used written words to test this Effect, although more recently faces have been used in conjunction with written words to make the Directed Forgetting Effect more generalizable and ecologically valid. With reference to the Mere Exposure research, familiarity of stimuli, even when presented unconsciously, has led to an increase in liking for those stimuli. The aim of the current study was to combine these two areas to determine if the Directed Forgetting Effect can be replicated with more complex and ecologically valid stimuli, namely videos, while determining whether the instruction to Remember or Forget the word presented in the video will affect the likeability ratings of the faces shown in the videos. The hypothesis was that there would be greater likeability of the faces associated with a Remember instruction compared to a Forget instruction, since Remember words would be more familiar, as verified through the Directed Forgetting Effect. Results of the study showed a very strong Directed Forgetting Effect of the videos, with higher recognition accuracy of Remember than Forget words. However, strong evidence was found against the difference between ratings for faces associated with Remember and Forget instructions.

T11) From gestures to gaming: Visible embodiment of one's cognitive state

Joseph Chisholm

The field of embodied cognition is based on the idea that cognition does not occur in isolation but operates within, and interacts with, a body. Consistent with this notion, research has demonstrated how bodily states can influence affective and perceptual processes. However, it is also clear that internal cognitive states can influence bodily states (e.g., to generate goal-directed actions); therefore, the observation of overt behaviour can be used to gain insight into one's cognitive state. Interestingly, we often engage in overt behaviour that does not appear deliberate or even appear to serve any clear purpose (e.g., gestures, leaning behaviour). A series of recent investigations have employed a natural observation approach to study these spontaneous behaviours in contexts ranging from a letter reading task to video game playing (teleoperation). I will present the results of this work and discuss how spontaneous behaviours can represent the visible embodiment of one's cognitive state.

T12) The Mediating Role of Gambling Motives in Explaining the Prospective Relations of Personality to Gambling Outcomes in Emerging Adults

Stewart, S. H., Mackinnon, S. P., Lambe, L., & Pearson, R.

We tested relations between the five-factor model traits of Neuroticism and Conscientiousness with gambling outcomes, as mediated through coping and enhancement gambling motives. Four-hundred-and-eighty-seven emerging adult gamblers (52.6% female; mean age = 22.2) participated as a part of the last two waves of the Manitoba Longitudinal Study of Young Adults (Manitoba Gaming Control Commission, 2011). Participants completed the NEO Five Factor Inventory (Costa & McCrae, 1992), the Gambling Motives Questionnaire (Stewart & Zack, 2008), and the Canadian Problem Gambling Index (CPGI; Ferris & Wynne, 2001). Multiple mediator analyses, using bootstrapping, revealed the hypothesized significant indirect effects of

low Conscientiousness on money spent, gambling frequency, and gambling problems mediated through enhancement motives, and significant indirect effects of high Neuroticism (expected) and low Conscientiousness (unexpected) on money spent gambling that were mediated through coping motives. Findings suggest two relatively distinct pathways to excessive/problematic gambling among emerging adults. Prevention implications will be discussed.

T13) Don't Stumble Tumble: A Videogame to Reduce the Use of General Anesthesia in Pediatric Magnetic Resonance Imaging

Melissa Howlett and Jill Chorney

Magnetic resonance imaging (MRI) is distressing for many children, as they must remain immobile and endure loud noises, while lying in a narrow, enclosed tube. Some children even require general anesthesia (GA) in order to complete the scan successfully. Currently, there is no effective and accessible preparation tool available for MRI; as such, this research aims to test the efficacy of a MRI-preparatory videogame for children. This videogame is designed to prepare children for the sensory experience of MRI, while also shaping skills to enhance their capability of completing the scan without GA. Beta testing will test the game with children, in order to identify functional issues and incorporate feedback into remodelling. Implementation testing will provide the opportunity for children referred for MRI with GA, to use the videogame and attempt MRI without GA, thereby determining the proportion of families willing to use the videogame and whether the videogame is successful in reducing the necessity of GA. This tool has the potential to increase healthcare efficiency by reducing the time and resources necessary to acquire quality scans, as well as provide children with an enhanced MRI experience.

T14) Real Time Neurofeedback During Imagined Imitation of Arm Movements: A Pilot Study

Christopher Friesen and Shaun Boe

While various forms of mental exercises to promote motor recovery—specifically Action Observation (AO) and Motor Imagery (MI)—can be performed at home and have been proven efficacious, at the moment they are rarely used. This pilot project will combine AO and MI, where an individual watches movements *whilst imagining they are executing those same movements*, while providing real-time neurofeedback via electroencephalography (EEG). This protocol of *Imagined Imitation* (II) has previously been shown to engage the motor network to a greater degree than either AO or MI alone. Furthermore, while MI with neurofeedback has been shown to more effectively engage the sensorimotor areas required to induce motor recovery from stroke than MI alone, this has not yet been attempted using II. In this pilot project healthy subjects will perform II of a unilateral upper-limb movement, while receiving information about the symmetry of their sensorimotor activity. This project will provide valuable information for those interested in designing an optimal at-home adjunct to physiotherapy to promote motor recovery following stroke.

T15) Exposure to tobacco-related cues does not enhance cravings in patients with psychotic disorders relative to control smokers

Marie-Eve Couture, Sean P. Barrett, Hera Schlagintweit, Phil Tibbo, Sherry H. Stewart, Heather Milliken, & Kimberley P. Good

Rates of tobacco smoking in patients with psychosis are remarkably high, and are accompanied by low rates of successful smoking cessation. Exposure to smoking-related cues is shown to reliably enhance cigarette cravings in dependent smokers recruited from the general population. In contrast, it remains unclear whether or smoking-related cues affect smokers diagnosed with psychotic disorders to the same degree as they do non-psychiatric populations. In addition, it is unknown whether the smoking motives of patients with psychosis differ from those of non-psychiatric smokers. A greater understanding of cue reactivity and reasons for smoking in smokers with psychosis may lead to enhanced interventions and increased rates of successful smoking cessation in this population. The present study compared the smoking motives, and the cigarette cravings of patients with psychotic disorders to those of healthy control participants after completion of a cue reactivity paradigm. Results revealed patients with psychotic disorders tended to be less sensitive to the cue-reactivity paradigm and suggests that patients may not be affected by tobacco cues to the same degree, if at all, as their healthy control counterparts. Further, comparison of smoking motives suggests no differences between the reasons for smoking reported by patients and those reported by healthy controls. Findings have implications for future studies aimed at identifying the source of the wide discrepancy in rates of tobacco smoking and smoking cessation across these two populations, and for the development of effective smoking cessation interventions for smokers with psychosis.

T16) A pilot study to examine the effects of computerized working memory training on working memory and attention in people with Parkinson's disease

Stephanie A. H. Jones, Elaine Skene, Janet Green, Gail A. Eskes

Working memory (WM) is the ability to actively hold and manipulate information in one's mind and is needed for many everyday tasks such as reading and following a conversation. More than half of people with Parkinson's disease (PD) exhibit cognitive impairment, including WM deficits (Kehagia et al. 2010). We examined whether using a WM computer activity benefits WM and attention in people with PD compared to a control group. PD participants were randomly assigned to one of two computer activity groups: WM or control (CT). Each activity was used five days per week (30 minutes/day) for five weeks. Age-matched healthy adults were randomly assigned to use one of the computerized activities. WM activity: an adaptive dual n-back activity in which participants had to monitor a series of visual and auditory stimuli and press computer mouse buttons when the current stimulus matched a previous stimulus 'n' items ago. The 'n' was adaptive in that it adjusted according to the performance of the individual. CT activity: participants moved and rotated blocks to form horizontal lines (non-adaptive). All participants completed a computerized battery of attention tasks before and after the five weeks of activity use. We examined baseline differences in reaction time and accuracy for each independent variable within each of the attention tasks. Hedges' g was used to quantify the change in reaction time and accuracy before and after computerized activity use in the WM and CT groups. Preliminary results indicate that at baseline, PD participants show attention and WM

deficits similar to previous reports (problems with initiation, inhibition, response interference, monitoring and WM). After training, the WM training group shows particular improvement on tasks requiring initiation, response interference and verbal WM compared to the control group.

T17) Building a cortical network of semantic concepts in a second language: ERP evidence of fast learning via computerized games

Kiera O'Neil, Aurélie Lagarrigue, Aaron J. Newman, Cheryl Frenck-Mestre

What does it take to acquire a semantic network in a second language? The present ERP study shows extremely rapid instantiation of both learned words and related concepts, via computerized games (“LANGA”). Participants served as their own control. Electrical activity of the brain, recorded at the scalp, was examined prior to exposure to the second language and 8 days later, following a 6 day training session (preceded and followed by orientation and consolidation, respectively). Participants learned 12 words per day (nouns and verbs), for a total of 72 words over 6 consecutive days. Results show rapid changes in cortical activity, associated with learning. Prior to exposure, no modulation of the N400 component was found as a function of the correct match vs. mismatch of audio presentation of words and their associated images. Post training, a large N400 effect was found for mismatch trials compared to correctly matched audio-visual trials. More importantly, images that were semantically related to learned words (eg. for the learned word “horse” the image of a saddle was presented), produced a reduction of the N400 compared to mismatched pairs (eg. the same learned word “horse” paired with the image of a building). This semantic priming effect suggests that the words in the new language were rapidly integrated with existing semantic networks, rather than existing as an isolated second language lexicon. Our results attest to the plasticity of adult learners’ brains and provide evidence for rapid onset of a semantic network in a late learned language.

T18) Virtual reality environments for aphasia rehabilitation

Francesco Usai and Aaron J. Newman

Current online applications for aphasia rehabilitation show high rates of attrition in the early stages of the training process, limiting the potential for optimal recovery. The main limit of these applications is a design overlooking important factors for a successful therapy, such as the level of immersion and motivation the users experience while doing the exercises. In this regard, this project aims at creating a cost-effective, easily accessible framework for aphasia rehabilitation that overcomes the limits of current online therapies. To achieve this goal, we will use principles of serious gaming and virtual reality to design evidence-based therapies that can engage and motivate the user to attain constant and intensive training required for a successful rehabilitation. Product design will be driven by evidence from relevant literature and basic research done in our lab, together with feedback from subjects with aphasia taking part to the process of validation.

T19) Diffusion Tensor Magnetic Resonance Imaging and Assessment of Olfactory Function as Early Indicators of Parkinson's Disease and Alzheimer's Disease
Rebecca Hastings & Dr. Kim Good

Alzheimer's Disease (AD) and Parkinson's Disease (PD) are the two most common neurodegenerative disorders amongst Canada's geriatric population. It has long been understood that olfactory impairments are manifested in both AD and PD and that these deficits are often present before the cardinal symptoms appear. A robust literature has shown that those afflicted with AD and PD perform poorly on the University of Pennsylvania Smell Identification Test (UPSIT) when compared to age-matched controls. The olfactory bulbs and tracts are shown to be abnormal when assessed using diffusion tensor magnetic imaging (DT-MRI), but only in PD patients. No study has yet combined objective olfactory ability with brain imaging techniques to compare those diagnosed with AD with those diagnosed with PD. The current study aimed to assess olfactory function in AD patients and PD patients along with a group of age-matched healthy controls. Olfactory function was assessed with the UPSIT, while DT-MRI techniques were used to examine the integrity of white matter in olfactory tracts and in the parahippocampal gyri. Results from the present study indicated that patients with AD and patients with PD suffer significant olfactory impairments relative to healthy controls. No significant differences were found in integrity of white matter tracts in either the olfactory bulbs or the parahippocampal gyri.

T20) Inducing Spike Timing Dependent Plasticity in the Human Brain Through Paired Associative Stimulation.

Tony Ingram and Shaun Boe

In the mid 20th century Hebb famously proposed that the timing of neuronal discharge underlies synaptic plasticity. This phenomenon is now referred to as spike-timing-dependent plasticity (STDP). Two forms of plasticity, long-term potentiation and depression, occurs when a postsynaptic neuron discharges within ~20 ms after the presynaptic neuron (potentiation) or when it discharges ~20-100 ms before (depression). Recent studies indicate that STDP-like changes can be induced similarly on a larger scale — between brain areas — in humans using a paradigm known as paired associative stimulation (PAS). PAS involves stimulating two areas of the brain in close succession, allowing for facilitation of either long-term potentiation or depression between the targeted brain areas. Thus PAS may prove to be a powerful tool in research and clinical practice by aiding in functions like learning and memory. This talk will outline the existing literature on this approach, and propose applications for facilitating motor learning using PAS.

T21) Sequential actions: why do upcoming perceptual - but not motor - tasks affect grasping?

Kevin LeBlanc & David Westwood

Past research has shown strong evidence supporting the notion that the visual control of action and the visual perception of objects are mediated by two functionally and anatomically distinct visual systems. Little is known about how each visual system interferes with the other when performing a sequential task. However, it is known that the kinematics of an action can be affected by a subsidiary attention task. In the current study participants were presented with two rectangular objects placed one in front of the other. Participants were instructed to grasp the first object and place it on a specified target area and then either grasp, make a perceptual judgment about, or ignore the second object. The results revealed that preparing an action to the second object does not produce interference to the first action, but attending to its size for verbal judgment does.

Specifically, as the size of the second object increased, the amplitude of peak grip aperture towards the first object also increased when performing the perception condition.

T22) Is there a pharmacological intervention to replace darkness? Preliminary results from a collaborative study (M.I.T) of the use of intraocular injection of tetrodotoxin (TTX) to recover vision in animal models of amblyopia

Donald Mitchell, Kevin Duffy, Paige Northrup, Ming-fai Fong and Mark Bear

We previously demonstrated in kittens that the profound deficit in vision following an early period of monocular deprivation (MD) can be blocked or eliminated by 10 days of complete darkness (Duffy & Mitchell Curr Biol. 2013). The loss of visually driven neural activity associated with darkness may reset the levels of certain plasticity braking molecules in the visual cortex to low levels typical of an earlier stage of development. The increased plasticity conferred by reduced levels of braking molecules following immersion in darkness may promote the fast recovery from the anatomical and behavioural effects of a prior period of MD. We have investigated the possible benefits of the use of intraocular injections of TTX, a potent marine neurotoxin that selectively binds sodium channels and temporarily blocks action potentials. We examined whether blockage of *all* retinal activity (not just visually driven activity) with intraocular TTX recapitulates the beneficial effects of darkness.

T23) Feedback content can alter ERPs and perceptual realignment during a prism adaptation task

Stephane MacLean, Yoko Ishigami, Cameron Hassall, Olav Krigolson, Gail Eskes

Prism adaptation (PA) is a demonstration of perceptual-motor plasticity, but is also a promising rehabilitation tool for visuo-spatial neglect (VSN). During PA, participants perform a visuomotor accuracy task while their visual field is shifted laterally by prism goggles. The visual shift initially causes large accuracy errors, but adaptation takes place after repeated trials. Upon returning to normal vision, participants show an aftereffect – a visuomotor bias in the opposite direction of the preceding prism shift. For VSN patients, it is important to develop PA paradigms that create robust aftereffects because these are what ameliorate their symptoms. Not all forms of “adaptation”, however, result in strong aftereffects. The present study investigated the effect of

presenting different forms of accuracy feedback to healthy young participants while they performed a PA task. We compared magnitude of aftereffects and event-related brain potentials evoked by onset of different feedback events. Results suggest that visual feedback of upper limb position at the end of each trial leads to stronger aftereffects compared to feedback that only indicates size of error on each trial. Visual feedback of upper limb also evokes a parietal ERP that may serve as a marker of perceptual “realignment” processes that are critical to producing strong aftereffects.

T24) Optogenetic silencing of mouse primary visual cortex affects contrast adaptation in V1 but not the LGN.

Jillian L. King, Kurt R. Stover, Matthew P. Lowe, Nathan A. Crowder

Contrast adaptation is induced by prolonged viewing of a high contrast gratings, and perceptually manifests as a decrease in perceived stimulus contrast. Neurons in the primary visual cortex (V1) show characteristic changes in their contrast response functions during contrast adaptation, but the cellular mechanisms behind this phenomenon are largely unknown. Our lab has previously measured contrast adaptation in mouse V1 and found close similarities to higher order mammals; therefore we have taken advantage of the genetic tools available in mice to probe the mechanisms of contrast adaptation. We performed extracellular recordings in both V1 and the lateral geniculate nucleus (LGN) in transgenic mice that express channelrhodopsin-2 (ChR2) in GABAergic neurons while optogenetically silencing V1 to investigate the role of spike rate on adaptation. We found that silencing the cortex while the adapting grating was present reduced the magnitude of adaptation in V1 cells, and also caused an increase in sensitivity to low contrasts. This latter effect had never before been observed in non-perturbed circuits. Adaptation changed the responses of LGN neurons very little except for an increase in sensitivity to low contrasts. These results have allowed us to estimate what proportion of contrast adaptation observed in V1 can be attributed to inheritance from LGN and intrinsic processes from the spiking of V1 neurons themselves. We hypothesize residual adaptation not accounted for by these two factors arises from synaptic depression at geniculocortical synapses.

T25) Horizontal cell inhibition of cones via ATP hydrolysis

Sunghan Jung and William H. Baldrige

Retinal horizontal cells (HCs) modulate the output from photoreceptors via inhibitory feedback. It has been suggested that feedback is due to changing pH in the synaptic cleft between photoreceptors and HCs. One model proposes that phosphate, produced by extracellular degradation of ATP released through pannexin-1 channels of HC dendrites, acidifies the synaptic space. To test this model we studied the effect of drugs that interfere with the proposed pathway on an indicator of feedback, the rollback of the HC response to light. The effect of the pannexin-1 blocker, probenecid, and the ecto-ATPase inhibitors, ARL 67156 and BGO-136, were tested. ARL 67156 reduced HC rollback but neither BGO-136 nor probenecid had a significant effect. The action of ARL 67156 on HC rollback is consistent with the ATPase/proton model of feedback. The lack of an effect of probenecid and BGO-136 on HC rollback is not consistent with the model suggesting that additional work is necessary to characterize feedback.

T26) Do zebra finches speak Russian? An operant task to test discriminability of mothers' song and speech to their infants.

Jordan Fisk, Christine Tsang, Simone Falk, Leslie Phillmore

Discrimination between human song and speech can be difficult when mothers are communicating with their infants because "motherese" is song-like in delivery. In previous studies, Tsang et al. used a preference test to demonstrate infants aged 6-9 months were able to discriminate between audio clips of mothers singing and mothers speaking to their babies in Russian. However, in a different study by the same authors, adults had difficulty discriminating some ambiguous English stimuli; they could not accurately categorize them as song or speech. In this project we used an operant task to test whether zebra finches could discriminate between audio clips of human mothers singing to their babies in Russian and English from clips of mothers speaking to their babies in Russian and English. We also presented birds with the ambiguous English stimuli to see how they categorized them. Finches were able to accurately discriminate song from speech in both languages.

T27) The dynamics of inhibition in the aftermath of a cue is contingent on reflexive oculomotor activation.

Ralph S. Redden, Matthew D. Hilchey & Raymond M. Klein

Inhibition of return (IOR) is considered an inhibitory aftermath of visuospatial orienting, expressed in the form of slower responding toward the locations of prior orienting responses. As shown by Taylor and Klein (2000), the form of IOR depends on the saccadic response demands imposed by the task and, hypothetically, whether the oculomotor response system for reflexively generated saccades is in an active state. To test this hypothesis, we evaluate IOR by requiring a manual discrimination response preceded by reflexive and non-reflexive eye movements in separate conditions. The findings suggest that the state of the oculomotor system determines the form of IOR.

T28) Three optical mechanisms in wing spots of a large dolichopodid fly, *Liancalus genualis*.

Steve Shaw

Most flies have plain or darkly mottled wings, but *Liancalus* males have wing spots that redirect and diffuse sunlight using ~1515 tiny upright hairs, forming a bright signal for conspecifics. When the sun is obscured, a second signal is still possible because the spots fluoresce bluish-white in UV light, an important component of skylight. There is much blue in skylight though, raising the question of whether the fluorescence could even be seen over reflection from the rest of the wing. Reflection, measured with collimated blue light was surprisingly low. Quantification of fluorescence required several steps, and indicated impressive efficiency: only 12 UV photons are required to displace one fluorescent photon from the wing spot. The wavelength distribution in 5000K skylight then dictates that the natural fluorescence-to-reflection ratio will be high, ~19:1. Fluorescent spots should be easy to see. Accordingly, photographs made under natural skylight revealed a spot visual contrast of ~15:1. However, when sky UV was selectively filtered out, spot brightness dropped by only 14-18%. This

exposes a third signaling mechanism: the spot's refractile hairs act as waveguides and pick up skylight but reflect about 17% back. Fluorescence scores at only ~1/6 of this.

T29) A Comparative Approach to Understanding Sickness Mediated Behavioural Changes
Ken Sullivan

Sick animals typically change their behaviour, becoming lethargic, disinterested in food, water and social activities and run a fever. This change in behaviour, termed sickness behaviour, is considered an adaptive response and is thought to enhance immune function. Sickness behaviour is observed not just in mammals, but in animals across all phyla, suggesting this change in behaviour is a highly conserved evolutionary adaptation. If this is true, then understanding how sickness behaviours have been adapted over time in different species is crucial in understanding and controlling its positive (and negative) influence. A great deal of previous work has focused on studying sickness behaviour in mammals. We have used the Texas field cricket *Gryllus texensis* to study sickness mediated behavioural changes in order to fill in the gaps in our understanding of the adaptive nature of sickness behaviour.

T30) Coping with constraints: nutritional stress elicits changes in immune function in the caterpillar, *Manduca sexta*
Kurtis Turnbull, Ilya F. Kovalko, Gillian Davies, Russell H. Easy and Shelley A. Adamo

It is thought that nutrition may reconfigure the immune system to optimize immune function when resources are constrained. However, few studies provide empirical evidence that immune resources are redistributed during nutritional limitation. We examined the effect of short-term nutritional stress on the immune function and energy availability of fifth instar *Manduca sexta* larvae. Insects reared on a low-nutrition diet (25% nutrient) for two days exhibited pathogen-specific declines in resistance, but did not show changes in energy availability, immune measures or immune-related gene expression. In contrast, the nutrition-absent diet (0% nutrient) caused declines in overall pathogen resistance, infection tolerance measures and energy availability. Surprisingly, the *M. sexta* larvae reared on the nutrition-absent diet upregulated both resistance mechanisms and immune-related gene expression. We hypothesize that the cost of inducible immune responses (i.e. immune resources synthesized in response to a pathogen) may increase during nutritional stress and therefore, constitutive immunity (i.e. immune resources present even when there is no infection) is upregulated. Together, these results suggest that assumptions about the costs and benefits of different immune components may not hold under nutritional constraints.

T31) Coping with stress early in life to build resilience for later: Further evidence for the negative effects of helicopter parenting.

Tara Perrot

Environmental enrichment (EE) produces a host of effects in mammals that are associated with a shift toward the resilient end of the vulnerable-resilient continuum that defines individual differences in the development of psychiatric conditions. The mechanism by which environmental enrichment, a non-drug and non-surgical manipulation, is associated with such robust effects remains speculative. One recent hypothesis states that EE is a form of inoculation stress – successfully coping with the chronic mild stress associated with EE produces protective phenotypes. Our unique model of EE in which mother rats are housed in enhanced home cages (EHC) with their young produces long-lasting changes in emotional and social development of the young. More recently, we have been investigating how the EHC alters maternal behaviour as one means for altering offspring phenotype. I will summarize our recent findings using the EHC and discuss them within the context of the inoculation stress hypothesis.

T32) A scoping review of information about child health shared over social media

Michelle E Tougas, Christine T Chambers, Penny Corkum.

Social media facilitates information exchange and is increasingly being used by professionals and general public. Little is known about child health and social media interaction.

Aim: 1) Adapt systematic review methodology to explore social media interaction about pediatric pain and pediatric sleep; 2) identify and categorize main themes, 3) assess user interactions across social media platforms, and 4) assess consistency of posts with research evidence. **Methods:** Twitter and YouTube will be searched for posts about pediatric pain and pediatric sleep. Themes will be categorized and extracted. Category frequencies and user metrics will be analyzed. Netlytic will be used to summarize and discover social networks. Post content will be compared to research to identify the quality of information shared on social media.

Results: This presentation will outline the novel methodology and planned stages of research that will be undertaken. **Conclusions:** This work will evaluate an approach for describing health-related social media, will examine how this information is being shared, what questions are being asked, and what information is being disseminated.

T33) Gender Differences in the Mutual Influences of Drinking Motives in Couples

Ivy-Lee Kehayes, Sean P. Mackinnon, Ken Leonard, Ron Fraser and Sherry H. Stewart

Drinking is related to various negative consequences (e.g., memory loss, injury). Drinking motives in the five-factor model (enhancement, conformity, social, coping-with-anxiety, and coping-with-depression) have been linked to different drinking patterns. Interpersonal factors are also important predictors of drinking, as couples enter into a “drinking partnership,” where one partner’s drinking influences the other’s. Additionally, gender influence models suggest men hold power over women and influence women more than vice versa. This project sought to integrate drinking motives, drinking partnerships, and gender influence models to explain how romantic relationships influence drinking motivations over time. One hundred heterosexual couples were assessed once a week for four weeks. It was hypothesized that motives would be influenced by both the individual and the partner, and that men would influence women more than the reverse. Such results would demonstrate the importance of gender influence in romantic relationships and their effects on drinking motives. Preliminary results will be shared.

T34) Relationship conflict and drinking to cope: An actor-partner interdependence model

Mackinnon, S. P., Pieterse, A., Keyhayes, I., Leonard, K., Fraser, R., & Stewart, S. H.

Past research on couples has demonstrated that relationship conflict is a risk factor for alcohol-related problems through drinking to cope (Lambe et al., submitted). It was theorized that conflict increases negative emotions (i.e., anxiety and depression) which in turn leads people to drink as a means of coping with those emotions. However, our past work did not incorporate symptoms of anxiety and depression. The present study tests whether anxious/depressed affect mediates the conflict-coping relationship using a 4-wave, 4-week longitudinal study of romantic dyads. A sample of 103 romantic dyads participated, all of whom were pre-screened to have consumed 12+ alcoholic drinks in the past year. Significant indirect effects were found, with anxious/depressed affect mediating the relationship between conflict and coping motives. Significant actor effects (i.e., own affect predicting own motives) but non-significant partner effects (i.e., own affect predicting partner’s motives) were found. Implications and future directions will be discussed.

T35) Periodic limb movements during sleep in children with Attention-Deficit/Hyperactivity Disorder: Baseline frequency and impact of psycho-stimulant medication.

Esmot A Begum, Amy Goodday, Kathy F. Spurr, Malgorzata Rajda, Penny Corkum

To determine the prevalence of periodic limb movements during sleep (PLMS) in children with attention-deficit/hyperactivity disorder (ADHD) and the impact of stimulant medication on PLMS, 25 medication naïve children with ADHD and 25 typically developing (TD) children, 6-12 years of age, were evaluated using overnight polysomnography (PSG) in three phases: baseline, stimulant treatment and placebo treatment. Compared to the TD children, children with ADHD had significantly higher rates of PLMS (ADHD=24%, TD=4.0%, $p<0.05$) with longer sleep latency ($p<0.05$). Children in the ADHD/PLMS group shifted their sleep stages more frequently than children in the ADHD/Non-PLMS group ($p<0.001$). Stimulant medication was not associated with PLMS; however, children in the ADHD/PLMS group showed significantly longer sleep latency and decreased sleep efficiency during stimulant treatment compared to the placebo. The result of this study reveals that PLMS in children with ADHD is associated with more severe sleep problems and more side-effects on medication.

T36) Attentional biases in problem and non-problem gamblers

Amanda Hudson and Sherry H. Stewart

Although attentional biases have been widely studied in substance use disorders, relatively few studies have looked at attentional biases in gamblers. The current study examined attentional biases for gambling stimuli, relative to emotional and neutral stimuli, in problem and non-problem gamblers ($n = 57$). To do so, we used a Rapid Serial Visual Presentation (RSVP) paradigm, which decomposed different components of attention, namely attentional capture and maintenance/ disengagement, by employing different time lags between distractors and targets. Problem gamblers showed attentional biases for gambling stimuli, relative to all other distractors, and these biases were observed only at the level of attentional maintenance/ disengagement. Non-problem gamblers showed some evidence of non-specific attentional biases (to both negative and gambling pictures). These findings highlight important differences in attentional processes between problem and non-problem gamblers, and suggest that sustained attentional biases among problem gamblers might be worthy targets for clinical interventions.

T37) Emotional Decision-Making and Psychotic-Like Experiences

Lynn MacKenzie, Rudolf Uher and Shannon Johnson

Background: Emotional executive functioning has previously been labelled hot executive function as this process involves an emotional component. Performance on emotional executive functioning has been linked to the ventromedial prefrontal cortex of the brain (Bechara et al, 2000). Hot or emotional executive function has yet to be explored in individuals at high risk for SMI and may yield valuable results to the field of research. Recent research has shown promising results for improving executive functioning abilities through early intervention (Diamond & Lee, 2011). Identifying an association between executive functioning deficits and psychotic-like experiences could provide useful information for clinicians selecting assessments and possible early intervention in high-risk populations. Exploratory analysis of emotional decision-making and psychotic-like experiences could also be meaningful to support further inquiry into testing feasibility and effectiveness of executive function interventions specific to this population.

Methods: Children and young people aged 8 to 21 will be selected from the existing Families Overcoming Risks Building Opportunities for Wellbeing cohort. The FORBOW is an accelerated cohort study enriched in risk of severe mental illness, which started recruiting participants in January 2013. FORBOW recruits the sons and daughters of people suffering from SMI and matched comparison controls. Overall, 75% of FORBOW participants have a family history of SMI in one or both parents, putting them at a 1-in-3 risk for developing a SMI themselves by early adulthood (Rasic et al., 2013). The remaining 25% are offspring of parents without SMI of similar age and form the same neighbourhood, referred to as comparison offspring (CO). FORBOW recruits participants at a young age (5 to 21) and follows them up in yearly intervals. The annual follow-ups include a comprehensive assessment of psychopathology and antecedents to SMI: cognitive impairment or delay, (Reichenberg et al., 2010) affective instability, (Egeland et al., 2012) anxiety, (Duffy et al., 2010; Kim-Cohen et al., 2003; Kovacs et al., 1989; Copeland et al., 2009) basic symptoms (Schimmelmann et al., 2013) and psychotic-like experiences. (Downs et al., 2013; Kelleher and Cannon, 2011; Polanczyk et al., 2010)

Implications and Directions: Since FORBOW is a prospective study, the scores on the emotional decision making test will further be used to predict emergence and persistence of PLE on follow-up assessments. The results from these investigations will inform the planning and design of early interventions. If impaired emotional decision making is associated with PLE, a training in emotional decision making may be included as part of interventions for young people with PLE.

T38) Why study the history of neuroscience?

Richard Brown

This presentation has 3 parts. First, what is history? Second, why study the history of neuroscience? In the third part, I will show some of the work we are doing at Oxford on our "Saving the history of neuroscience" project using examples from our history of neuroscience website [<https://history.medsci.ox.ac.uk/>]. The general purpose of the presentation is to show how history is embedded in each phase of scientific research and teaching.

T39) Representing the Brain: Modeling neuroanatomy from the Early Modern to the Contemporary period

Brianna Bird

This presentation will survey the history of anatomy modeling in reference to the brain and nervous system. Important developments include whole-body wax modeling in the 18th century, “moulages” as teaching models in the 19th century, and contemporary computer modeling. Given their visual nature, models engage non-experts and exert a strong influence on the popular imagination. Neuroanatomical models therefore reveal the tenor of broader ideas about the brain and the mind.

T40) Golgi versus Cajal and the Neuron Doctrine

Emre Fertan

With Robert Hooke's (1665) discovery of cells using his new microscope, biologists were able to study cells as the building blocks of living organisms. This led to the development of the cell theory by Matthias Schleiden and Theodor Schwann (1838). Due to the difficulties of studying neurons, they were excluded from the cell theory. The invention of the microtome (1770) and the discovery of the tissue fixing properties of formaldehyde (1859) enabled neuroanatomists to prepare brain tissue slices to study under the microscope. The turning point for cytoarchitecture was the discovery and usage of different dyes to stain the slides. Golgi developed his famous staining technique using silver and made neuronal bodies and extensions visible, but he could not identify the small gaps between the neurons. Santiago Ramon y Cajal, modified Golgi's method to stain different nerve tissues and defined the neuronal doctrine, which states that neurons do not form continuous webs, but are separated from each other by small gaps. This led to the famous arguments between Golgi and Cajal when they were jointly awarded the Nobel Prize in 1906.

T41) Wilder Penfield's lasting contributions to neuroscience

Ryan Wheeler

Dr. Wilder Penfield is perhaps the most well known Canadian neuroscientist. Penfield's surgical work on epilepsy was ground-breaking. However, it is his work on cerebral localisation which put his name into introductory psychology textbooks across the globe. What is often forgotten, is his incredible educational experience as a Rhode's scholar, working with greats like Sir William Osler and Sir Charles Sherrington, to name a few. As founder of the Montreal Neurological Institute, Penfield played a pivotal role in creating a world famous destination for neuroscience research in Canada. Penfield's brilliant work, and the creation of the MNI, has influenced many other famous Canadian neuroscientists such as Donald Hebb and Brenda Milner. His influence not only lives on in the famous homunculus drawing but also in the long lasting impression he has made on neuroscience research in Canada and the world.

T42) I'm significant and you're not
John Christie

The title may not sound meaningful to you because, alone, it's not. But worse, if I relied on it alone to argue about meaningful differences I would be making a serious error. This talk was inspired by watching our honours conference and how our students are trained to interpret statistical tests.

Special Adjunct Talks

T43) Do children with dyslexia have deficits in spoken sentence comprehension?
Erin Robertson

POSTERS

Thursday's Posters

P1) Looking at endogenous prior entry effects in baseball judgements at first base

Ghislain d'Entremont, RJ Redden, Mike Lawrence & Raymond Klein

To examine whether Tichener's law of prior entry holds in a baseball context, participants were presented with close plays at first base. After each play, participants made a "Safe" or "Out" judgement, or reported the perceived colour of a disk flashed briefly over either the base or the glove. The biased presentation of coloured disks at one location or the other within a block was intended to endogenously orient attention towards that location. A novel colour wheel analysis was implemented and compared with the conventional analysis. The attentional manipulation was successful, but there was no visual prior entry effect.

Poster 2 moved to Friday

P2) Exploring sleep and circadian rhythm disturbance in an animal model of early stage Alzheimer's disease

Roxanne Sterniczuk, Setareh Lahasee and Ben Rusak

Sleep disturbance and impairment to the sleep-wake cycle, are commonly reported by individuals in the early stages of Alzheimer's disease (AD). Using a triple transgenic mouse model of AD (i.e., 3xTg-AD), this study examined whether there are alterations to the circadian pacemaker, the suprachiasmatic nucleus (SCN), prior to the development of the hallmark neuropathological changes that are characteristic of the disease (i.e., plaques and tangles). Previous work of ours found that 3xTg-AD mice show changes in neuropeptide profiles in the SCN when the brain exhibits AD neuropathology. Prior to the development of AD neuropathology however, there are already alterations in the activity/rest cycles of these mice. It is not known at what stage of disease progression changes within the SCN occur. Preliminary results indicate decreased clock gene protein product Period1 in the SCN prior to neuropathology. Findings from this study may help to identify novel biomarkers that could lead to the earlier and more accurate diagnosis of AD

P3) Investigation of glial fibrillary acidic protein (GFAP) in the primary visual cortex during normal development and in the absence of visually-driven activity.

NESC3370: R. Kindell, J. Morrow, A. Shahisavandi, H. Thorpe, K. Trotman, A. Hidayat, M. Barbaro, J. Blacklaws, A. Brown, J. Greenan, S. Kwiatkowski, L. Meisner, A. Pieterse, S. Shehzaad, M. Taylor, JA Horne, Kevin Duffy

Neurons within the mammalian primary visual pathway exhibit a high capacity for neural plasticity early in postnatal development. This critical period of neural development eventually closes so that little plasticity remains in adolescence and adulthood. The natural reduction of plasticity that occurs with age appears to be partly rooted in the accumulation of proteins – called molecular brakes - that stabilize neural circuits refined by early visual experience. Astrocytes are among the most plentiful cells within the central nervous system and they have been implicated as facilitators of neural plasticity. In this study we examined a principal constituent of the astrocyte cytoskeleton, glial fibrillary acidic protein (GFAP), to determine if it is developmentally regulated within the visual cortex in a manner consistent with a molecular brake. In the second part of our study we examined whether dark-rearing, a condition known to enhance neural plasticity, alters GFAP labeling in visual cortex. While GFAP levels appeared not to be developmentally regulated within the developing visual cortex, labeling for GFAP was altered when visually-driven activity was removed through dark-rearing.

P4) Age related changes in anxiety, motor performance and spatial learning in the 5xFAD mouse model of Alzheimer's Disease.

Emre Fertan and Dr. Richard E. Brown

We used the Triple Test of Anxiety (TTA), Rotarod and the Hebb-Williams (HWM) maze to assess anxiety, motor performance and spatial learning and memory in 5xFAD mice at 6 and 12 months of age. We observed less time spent in the closed arms of the elevated plus maze for the 5xFAD mice, yet they spent more time in the dark box. We found motor deficits in the 12 month old 5xFAD mice on the Rotarod which affected both their latencies and motor learning. While there were no significant differences between the WT and 5xFAD mice in the acquisition phase of the HWM, all the mice had longer latencies and higher number of errors in the hard mazes of the test phase and there was a trend of 5xFAD mice making more errors compared to the WT controls in all three levels of difficulty.

Friday's Posters

P5) Using the SpyTrain videogame as a means to improve visual search in the older adults

Michael Gardner and Ella Daly

The capacity for our brains to perceive information and generate knowledge is known as cognition. With age and the onset of health problems associated with age, this ability gradually begins to decline. Videogames have shown some promising effects on cognition in past studies, so the “SpyTrain” game was created to train visual search skills. We used SpyTrain to measure the effect of age, videogame and computer experience on reaction time and accuracy. Fifteen older and fifteen younger adults participated, and were asked about their weekly videogame and computer use. These participants then individually played “SpyTrain” while their reaction time and accuracy was being monitored. We found strong correlations between age and reaction time in the old group and computer use and error in the young group, suggesting that visual search declines with age and that frequent computer use can improve accuracy. Trends were also found between age and error and computer use and reaction time in the young group, implying the older participants might be putting in more effort and those who often use computers are sacrificing time to reduce error. We were unable to find any correlations with videogame use due to a lack of gamer participants and we found no correlation between computer use in the old group, suggesting seniors do not often use computers in an engaging way.

P6) Examining mood, drinking motives, and alcohol use over the menstrual cycle

Megan Hodgin, Amanda Hudson, Roisin M. O'Connor, Sherry H. Stewart

Alcohol use tends to vary over women's menstrual cycles, with increased intake occurring (pre)menstrually – a phase also characterized by enhanced negative mood. Together, these findings suggest that women may drink for particular emotion-focused reasons at specific points in their cycles. In the current study, normally-cycling women ($M_{\text{age}} = 22.58$ years, $SD_{\text{age}} = 3.43$) completed daily diary measures (via Smartphone surveys), with questions on mood, drinking motives, and drinks consumed each day for the course of a full cycle. Results thus far ($n = 26$) indicate that women felt less positive during menstruation and drank most heavily in the ovulatory phase, followed by menstrual and premenstrual phases, and lastly by follicular and luteal phases. Motives also varied by cycle phase, with women endorsing enhancement motives most in the ovulatory phase, and coping motives most (pre)menstrually. Social motives were commonly reported in both premenstrual and ovulatory phases.

P7) Anatomy and Pathology of the Human Olfactory Bulb in Alzheimer's Disease

H. Hamodat, M.K. Cash, S. Darvesh

Background: Clinically, there is evidence for olfactory dysfunction in neurodegenerative disorders, like Alzheimer's disease (AD), preceding typical cognitive and behavioural symptoms. In AD, olfactory impairment may result from neuronal loss, neuroinflammation and pathological deposition of β -amyloid ($A\beta$) plaques and tau neurofibrillary tangles (NFTs) in the olfactory bulb (OB). However, the laminar organization of the human OB is poorly understood and observations of AD pathology have been inconsistent. **Purpose:** the goal of this study was to discern the laminar organization of the human OB and determine structural and pathological changes in the AD OB. **Methods:** AD and normal control OB tissues were obtained from the Maritime Brain Tissue Bank in Halifax, Canada. Adjacent OB tissues were stained with Nissl to examine laminar distribution and immunohistochemical techniques to examine neuronal, AD pathological and neuroinflammatory markers. **Results:** Herein, we provide a detailed map of the anatomical distribution of the human OB. $A\beta$ plaques and tau NFTs were observed in specific layers of the AD OB, predominately in the anterior olfactory nucleus. A lack of pathology was noted in the olfactory tract. A neuroinflammatory response was observed through the presence of activated microglia, particularly in regions with noted AD pathology. **Conclusions:** The specific pattern of AD pathological changes observed in the AD OB, provides support that impaired olfactory functions may be related to the peripheral olfactory system. Further investigation of the AD OB is warranted for the potential use of olfactory dysfunction as an early, definitive biomarker and diagnostic target for AD diagnosis.

P8) Amyloid-beta plaque load, cell density, and soma size in the 3xTg-AD mouse model of Alzheimer's disease

Kylie Duncan, Miranda Frison, Michelle Lehman, Hannah Wood

Alzheimer's disease (AD) leads to the deterioration of thinking, memory, and motor function. The triple transgenic mouse model of AD (3xTg-AD) mimics the spatial and temporal progression of AD's neuropathological hallmarks – amyloid-beta ($A\beta$) plaques and neurofibrillary tangles. Previous studies found that 3xTg-AD mice showed superior motor coordination relative to controls (B6129SF2/J) on the rotarod apparatus. Given that motor deficits in human AD arise from $A\beta$ plaque-induced neurotoxicity in the primary motor cortex (M1), the current study was conducted to determine if a relationship exists between $A\beta$ plaque load, soma size, and cell density in layer V of M1 of 3xTg-AD mice. We used ImageJ to quantify $A\beta$ plaque load, soma size, and cell density in two- and 15-month-old male and female 3xTg-AD and control mice. We found that 3xTg-AD females expressed significantly higher plaque loads than genotype- and age-matched males at both ages. Although a main effect of genotype on cell density or soma size was not observed, a main effect of sex on soma size was found, with females showing larger cells than males, and a main effect of age on cell density was found, with two-month-old mice having higher cell density than 15-month-old mice. Plaque load was significantly correlated with soma size and cell density. These results suggest that further examination of the motor system of 3xTg-AD mice may uncover a neuroanatomical or neurophysiological mechanism that preserves their motor functioning, which could have implications for the treatment of motor disturbances in human AD.